

# RUTU MAHESH GHATGE

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## Summary

AI Engineer and AI & Data Science undergraduate with 2+ years of hands-on experience in Python backend development, computer vision, and NLP, building real-world AI systems using LLMs, CNNs, LSTMs, Kalman Filters, and YOLO. Internship experience across DRDO, Infosys Springboard, and AI startups, with strong skills in Python, TensorFlow, LangChain, and RAG architectures, along with exposure to IoT-based intelligent systems and published research in robotics.

## Skills

**Programming:** Python, C++, MySQL, MongoDB, SQLite3, Basic Assembly Language

**Libraries & Frameworks:** TensorFlow, PyTorch, OpenCV, Keras, LangChain, LangGraph, Scikit-learn, Pandas, NumPy, Matplotlib, Filterpy, BeautifulSoup, Selenium

**AI/ML Expertise:** CNN, LSTM, YOLO, DeepSORT, Kalman Filters, NLP, RAG, LLM Integration, Prompt Engineering

**Tools & Platforms:** Dialogflow CX, Vertex AI, React, FastAPI, Flask, Docker, Git & GitHub, Power BI, Microsoft Excel, AWS, Hostinger

**Soft Skills:** Team Leadership, Cross-functional Collaboration, Public Speaking, Problem Solving, Agile & Scrum, Technical Communication, Elocution

## Experience

### AI Engineer

June 2025 – Present

*Techs & Tomes (Skin Beyond Borders)*

*Pune*

- Deployed a multilingual skincare chatbot using LangChain and RAG, reducing average query resolution time by ~40%.
- Led complete backend development for the live Formulynx.ai project — designed APIs, database architecture, and AI integration to support real-time formulation analysis and deployment.
- Built AI video translator and summarizer using faster-whisper and Edge TTS, processing 50+ videos with 95%+ transcription accuracy.
- Created scalable web scrapers for skincare data extraction and cleaning, collecting 10,000+ structured product records.
- Designed Skin Vision, a privacy-preserving AI facial skin analyzer with budget-aware recommendations, integrated via React and Tailwind.
- Handled frontend for admin panel and website redevelopment of SkinBB, including API integration and UI implementation under tight deadlines using React, Tailwind CSS, and AI tools.

### AI Intern

Nov 2024 – May 2025

*ARDE DRDO*

*Pune*

- Designed a custom CNN model for aerial object detection using GELAN activation and KMeans-based anchor box optimization for high accuracy on small moving targets.
- Developed a hybrid Kalman Filter + LSTM model for real-time drone tracking with 99%+ predictive accuracy.
- Built a PDF-based RAG chatbot integrating LLaMA and DeepSeek to answer technical queries from 500+ defense documents.
- Researched autonomous drone-following-drone navigation using YOLO, DeepSORT, SLAM, and D\* path planning in dynamic environments.

### Smart Traffic Management Intern

Feb 2025 – Mar 2025

*Infosys Springboard*

*Remote*

- Developed ANPR and ATCC systems using YOLO, OCR, Python, Flask, and React with helmet detection, heatmaps, crosswalk and signal violation detection.
- Collaborated in an Agile environment with GitHub for version control, delivering all milestones within a 6-week sprint cycle.

### Development Intern

June 2024 – Nov 2024

*Standard Learner*

*Remote*

- Updated and maintained a mock test platform serving active student users.
- Implemented new features, resolved production issues, and collaborated on chatbot development and AI model integration.

## Data Science Intern

May 2024 – Jul 2024

*CodeClause*

*Remote*

- Worked on data science pipelines including EDA, feature engineering, and predictive model building using Python and Scikit-learn.

## Web Development Intern

Feb 2024 – Mar 2024

*TechnoHacks Edutech Ltd.*

*Remote*

- Built and maintained web application features using standard frontend and backend development practices.

## Business Analyst Training

Jan 2024 – Feb 2024

*Finlatics*

*Remote*

- Completed structured business analyst training covering data interpretation, reporting, and analytical decision-making frameworks.

## Projects

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### RahiAI – Intelligent Tourism Chatbot

- Architected a full-stack tourism chatbot providing personalized travel recommendations and itinerary planning by integrating real-time data APIs.
- **Tech Stack:** React.js · Python · FastAPI · LangGraph · MongoDB

### SafeSurf AI – Malicious Website Detector

- Engineered a browser extension providing real-time protection against malicious websites using a Scikit-learn classifier integrated with VirusTotal API, achieving 94%+ classification accuracy.
- **Tech Stack:** Django · Python · Scikit-learn · VirusTotal API · Selenium · Playwright

### Human Emulated Robotic Hand

- Developed an IoT flex sensor glove that transmits flex data to replicate hand movements on a robotic replica; published in IJARSCT and presented at ICMETET.
- **Tech Stack:** IoT · Flex Sensors · Arduino · Excel

### Multiformat Deepfake Detection

- Built a multimodal deepfake classifier for image, voice, and text with 91%+ accuracy using React frontend and Flask backend.
- **Tech Stack:** React · Flask · Python · Deep Learning

### Heart Disease Risk Assessment

- Constructed a GUI-based cardiovascular risk assessment tool using SQLite3 and PySimpleGUI, evaluating risk based on 10+ clinical input parameters.
- **Tech Stack:** Python · SQLite3 · PySimpleGUI

### AI/NLP Mini Projects

- Built NLP prediction models, sentiment analysis with K-means clustering, and text classification using NLTK/SpaCy and Scikit-learn; presented Deep Reinforcement Learning concepts and applications.

### Clap Switch – IoT Home Automation

- Designed sound-sensing electronic circuits for home automation using IoT and embedded electronics principles.

## Research Work

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### Human Emulated Robotic Hand

*Published – IJARSCT · Presented – ICMETET*

- Research on affordable neuro-prosthetics using a flex-sensor glove and micro-actuated robotic replica to emulate human hand movements.

### Multi-Criteria Decision Making in Mechanical Systems

*Ongoing*

- Applying Fuzzy-DEMATEL, VIKOR, and TOPSIS for performance evaluation and raw material supplier optimization in collaboration with Mechanical Engineering faculty.

### Early Prediction of PCOD/PCOS using Multimodal AI

*Ongoing*

- Developing a machine learning and computer vision approach combining clinical parameters, ultrasound features, and patient history for early risk detection using multimodal fusion and bias-aware medical ML pipelines.

### Theory of Mind in Multi-Agent Reinforcement Learning

*Ongoing*

- Designing ToM-based cognitive agents capable of modeling opponent beliefs and intentions in MARL environments, including ToM-net, opponent modeling, and cooperative behavior emergence.

Certifications

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- **Google AI Essentials** (Google, 2024)
- **Introduction to Generative AI** (Google, 2024)
- **Badge & Webhook** (Google Cloud Skills Boost, 2024)
- **SQL and Relational Databases** (IBM, 2024)
- **Foundations of Cybersecurity** (Google, 2024)
- **Infosys Springboard:** Mastering IoT with Arduino · Introduction to AI · Introduction to Deep Learning · 45+ additional certificates

Achievements

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- Winner – 1st place, DATAthon at ISBM COE
- Winner – 1st place, State-level Hackathon at Ajeenkya DY Patil School of Technology, Pune
- 1st Position – Project Competition, ISB&M Bangalore
- 1st Position – Idea Hackathon, ISBM Kolkata
- Top 5% with Elite + Silver Medal – NPTEL Mobile Virtual Reality and Artificial Intelligence Certification
- Second Runner-up – EDathon at Ajinkya DY Patil School of Technology, Pune
- Top 8 – Inceptia National-Level Hackathon at PCCOER, Pune
- Debate winner at multiple intercollegiate events

Extracurricular Activities

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- Coordinator, IoT Tech-lab and Research Centre
- Member, Technical Placement Cell ISBM COE; Member, Coding Club; Student Association Anchor and Treasurer
- Mentee, Infosys Springboard Pragati Program
- Active participant in anchoring, elocution, and debating events

Education

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**B.E. Artificial Intelligence and Data Science**  
*ISBM College of Engineering, Savitribai Phule Pune University, Pune*

**2022 – 2026**  
*SGPA: 8.62 / 10*

**HSC**  
*Loyola Junior College, Pune*

**2022**  
*Percentage: 73.33%*

**SSC**  
*Modern High School, NCL Campus, Pune*

**2020**  
*Percentage: 93.40%*

Languages

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English | Hindi | Marathi